

Online Fraud Payment Detection System

Project Title:	Online Fraud Payment Detection Using Balanced ML Algorithms
Core Engine:	Python 3.7.0 (Stable Environment Track)
GitHub Repo:	github.com/kanasanidhanush7-dev/online-fraud-payment-detection-using-ml-algorithms

1. Verified Production Requirements

The following core packages are locked and configured in your `requirements.txt` file. This minimal configuration guarantees clean compilation, eliminating previous syntax mismatch errors:

```
django==3.2
numpy==1.21.6
pandas==1.1.5
scikit-learn==0.24.2
```

- **django==3.2:** Manages the web server architecture, dashboard routing, data submission views, and interface templates cleanly.
- **numpy & pandas:** Structured multi-dimensional data array processing engine optimized to parse heavy financial transaction datasets.
- **scikit-learn==0.24.2:** Houses your primary machine learning algorithm pipelines, dataset splits, evaluation metrics, and prediction arrays.

2. Step-by-Step Environment Execution Setup

Step 2.1: Local Package Installation

To rebuild or sync your system dependencies from your project directory root, clear conflicting environments and execute:

```
pip install -r requirements.txt
```

Step 2.2: Local Development Server Deployment

To initiate your backend system dashboard routing pipelines and launch the predictive dashboard web interface locally, run:

```
python manage.py runserver
```

Once deployed, access your system dashboard via your local browser environment at: <http://127.0.0.1:8000/>

*To safely terminate the running engine session at any point, press **Ctrl + C** inside your command window.*

Step 2.3: Cloud Version Control and Backup

To back up any structural script alterations or modeling parameter files safely into your secured remote cloud repository tracking setup, execute the sequential sequence:

```
git add .
```

```
git commit -m "updated fraud detection backend logic"
```

```
git push
```